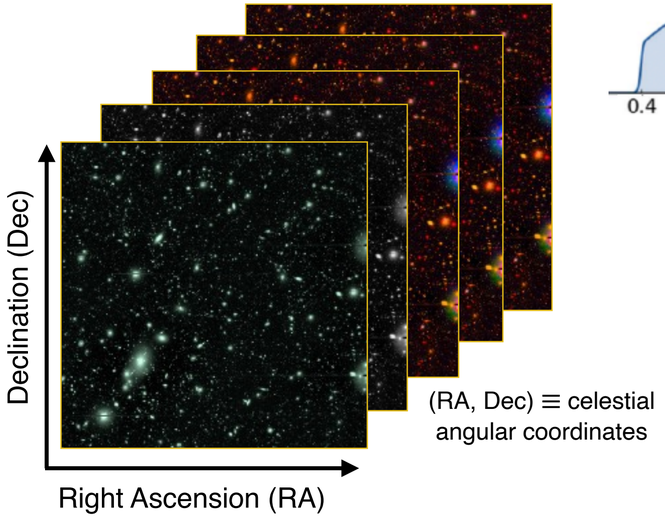
Bayesian modelling of galaxy populations

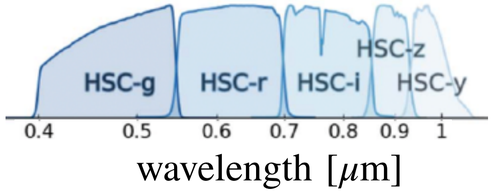
Focus on galaxy stellar-mass distribution in galaxy groups

Photometric observations and galaxy models

Observations



Typical filter transmission curves

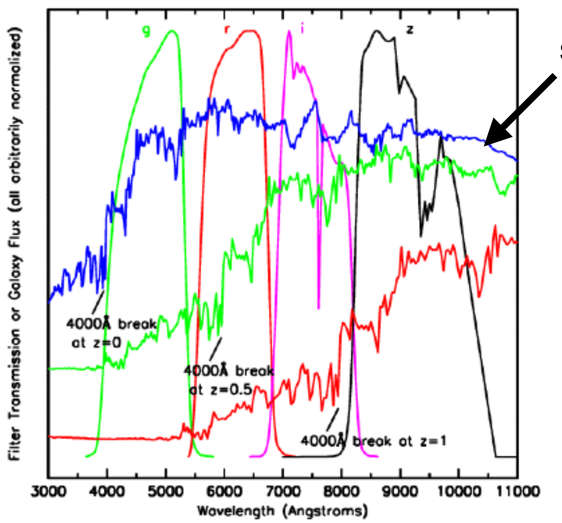


- We have images in multiple photometric large bands from ground based and space-based telescopes

- Using source detection softwares, we build catalogues of galaxies and measure their fluxes in each band in fixed angular apertures.

$\rightarrow \quad \text{data} \equiv \overrightarrow{\text{fluxes}}$ for each galaxy

SED fitting



Expected spectral energy distribution (SED) for a model galaxy

- Using physically motivated galaxy models, we can compute the expected observed fluxes for galaxies with known intrinsic properties $\overrightarrow{\psi}_{\text{phys}}$ (distance, stellar-mass $M_\star$, star-formation rate, etc.)

➔ We can thus solve the inverse problem in a bayesian framework through so-called *SED fitting* and get

$$p(\overrightarrow{\psi}_{\text{phys}} \,|\, \text{data}) \propto p(\text{data} \,|\, \overrightarrow{\psi}_{\text{phys}})\; p(\overrightarrow{\psi}_{\text{phys}})$$

➔ In practice, my latest work focused on $p(M_\star \,|\, \overrightarrow{\text{fluxes}})$